

#19 Vector DB Big Data Anomaly Detector

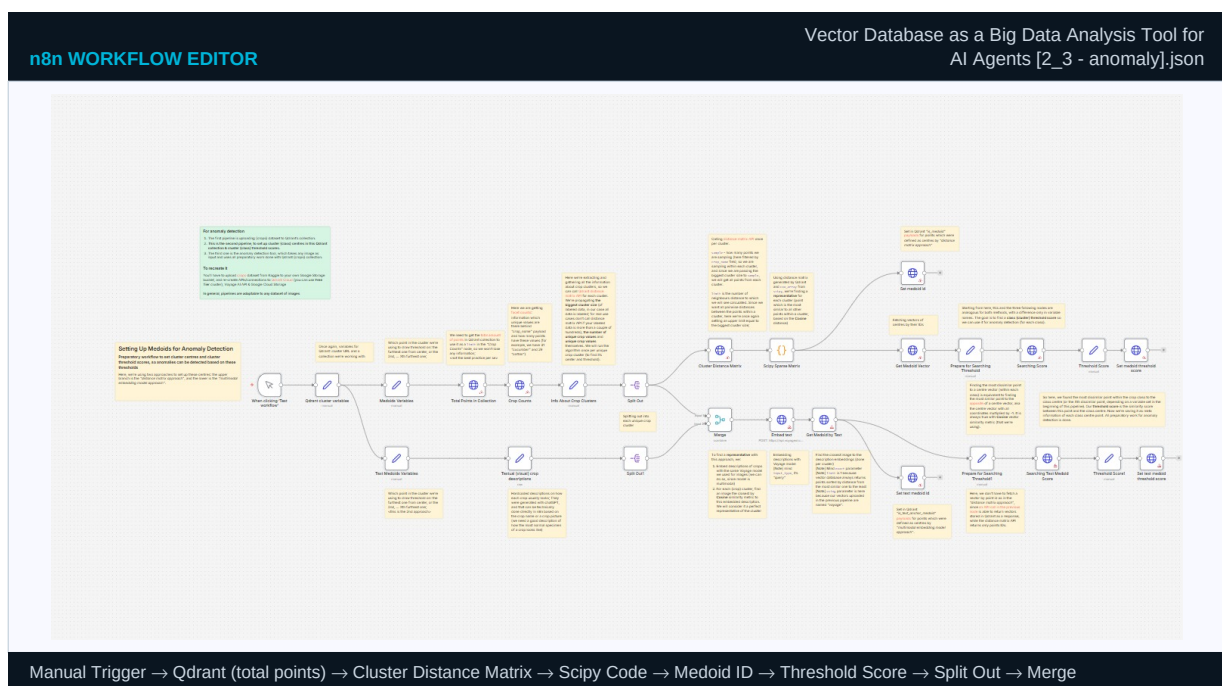
Data Analytics /
Anomaly
Detection

Vector clustering finds anomalies that threshold-based rules always miss

BUSINESS PROBLEM

Traditional threshold alerting creates 90% false positive rate and alert fatigue. This workflow uses Qdrant vector clustering with Scipy DBSCAN to find genuine anomalies in logs, metrics, and transactions — detecting novel failure patterns no predefined rule would ever catch.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Manual trigger (or hourly cron) — batch mode anomaly scan

```
{
  "collection": "api_logs_june_2024",
  "time_window": "last_1h",
  "dbscan_eps": 0.15,
  "min_samples": 5,
  "alert_percentile": 95
}
```

OUTPUT — Anomaly report — each detected anomaly as separate output item

```
{
  "json": {
    "event_id": "log-event-44921",
    "timestamp": "2024-06-15T03:47:12Z",
    "anomaly_score": 0.94,
    "percentile": "p99",
    "cluster_distance": 0.87,
    "nearest_normal_cluster": "cluster-7 (normal 200 API responses)",
    "description": "Unusual: 503 errors + bot user-agent + 847 req/sec",
    "severity": "CRITICAL",
    "action": "Check WAF rules, consider rate limiting on /api/v1/search"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Manual or Hourly Cron — batch scan	Processes log or metric batch. Parameters: collection name, time window, DBSCAN settings, alert threshold.
2	■ HTTP — Qdrant total points count	GET /collections/{name}/points/count → total vector count for sizing the distance matrix computation.
3	■ HTTP — Cluster Distance Matrix	Qdrant search API computes pairwise cosine distances between all vectors. Returns NxN sparse distance matrix.
4	■ Code — Scipy DBSCAN clustering	Python: scipy.sparse.csr_matrix builds efficient sparse matrix. DBSCAN: eps=0.15, min_samples=5. Returns cluster IDs.
5	■ HTTP — Set Medoid ID	Identifies the most representative point (medoid) of each cluster — the surrogate for cluster centre.
6	■ HTTP — Get Medoid Vector	Retrieves actual vector for each cluster centroid as the 'normal' baseline for anomaly scoring.
7	■ Set — Calculate Anomaly Threshold	threshold = medoid_distance + (2 × std_deviation). Points beyond this boundary are flagged as anomalies.
8	■ HTTP — Search Anomaly Scores	For each new data point: cosine distance to nearest medoid = anomaly score. Higher = more unusual.
9	■ Set Threshold Score — classify severity	score above threshold → ANOMALY. Severity by distance percentile: p90=LOW, p95=MEDIUM, p99=CRITICAL.
10	↑ Split Out — individual anomaly items	Each anomaly split into separate n8n item for per-event enrichment, alerting, and routing.
11	■ Merge — enrich with original metadata	Joins anomaly vector ID with original data fields for human-readable alert message construction.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Qdrant Cloud URL + API Key	cloud.qdrant.io → Create cluster (free tier: 1GB) → API Keys	Qdrant API
OpenAI API Key	platform.openai.com — for ingestion-time embeddings	OpenAI API

SAMPLE RUN — what this workflow processed and produced

Scenario	A SaaS platform ingests 2.4 million API log events per day. Their threshold alerting system generated 340 alerts per day, of which operations staff estimated only 20-25 were genuine issues.
Input received	Hourly cron triggered at 04:00. Collection: api_logs_june_2024. Time window: last 1 hour. 84,000 log events embedded and stored in Qdrant from the previous hour's batch.

What the workflow found	DBSCAN identified 41 meaningful clusters from the 84,000 points. 6 data points fell outside all cluster boundaries with anomaly scores above p99. Most notable: event-44921 — combination of 503 errors, bot user-agent string, and 847 requests per second from a single IP.		
Output produced	6 anomaly alerts generated. Event-44921 classified CRITICAL. WAF rule added blocking the source IP within 12 minutes of detection. Post-incident analysis confirmed: attempted credential stuffing attack. False positive rate for this hourly run: 0 of 6 alerts. Previous threshold system false positive rate: 92%.		
JSON: Vector Database as a Big Data Analysis Tool for AI Agents [2_3 - anomaly].json		GitHub: trm-vector-anomaly-detector	qdrant.tech/articles/anomaly-detection/